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MTL/NDT/2026

**REPORT ON AN IN-SITU INTEGRITY TEST
(NON-DESTRUCTIVE) OF COMPRESSIVE
STRENGTH OF STRUCTURAL ELEMENTS**

BY

**LAGOS STATE MATERIALS TESTING LABORATORY
OJODU BERGER, LAGOS.**

ON

Botanical Garden Road, Ebute Metta

FOR

-

Botanical Garden Road, Ebute Metta, Lagos, Lagos Mainland, Lagos

DATE OF TEST: NOT RECORDED

DATE OF REPORT: 05/09/2026

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EXECUTIVE SUMMARY

In situ Integrity Test (Non-Destructive) of compressive strength of structural members of the project "Botanical Garden Road, Ebute Metta", at Botanical Garden Road, Ebute Metta, Lagos, Lagos Mainland, Lagos.

The Non-Destructive Integrity Test was carried out on dates not recorded with the intention to determine the residual compressive strength of concrete component of the structural members considered to be critical to stability, robustness and general safety of the entire structure in its present state.

The visual inspection recorded 5 observation(s) on the tested structure (see Section 4.1), including defects such as cracks, spalling or dampness where noted.

The Non-Destructive test analysis shows that 0 of the 2 structural member(s) tested were good in strength (average compressive strength at or above the assumed 25 N/mm²), while 1 member(s) fell below it and require technical advice.

Members were tested on: FLOOR NOT RECORDED, Second Floor.

In view of the above, it is advised that a qualified structural engineer should be engaged to proffer solution to the defects observed, give technical advice on the poor structural members tested and further analyse the structural arrangement to guarantee the stability, integrity and the serviceability of the structure.

1.0 INTRODUCTION

Non-Destructive Test (NDT), as the name implies, means that the material under test is not damaged during test. Direct measurement of the strength of concrete involves destructive stresses and cannot be used for determining the quality of already cast concrete. It is for this reason that direct methods are not employed in determining the strength of in-situ concrete. As a result, indirect method was used.

In compliance with the Mandatory Non-Destructive Test requirement of the Lagos State Government, a Non-Destructive compressive strength test (Structural Integrity Test) was conducted on the project "Botanical Garden Road, Ebute Metta" located at Botanical Garden Road, Ebute Metta, Lagos, Lagos Mainland, Lagos. The map showing the exact location of the site is on the Location Map page of this report.

No structural drawing was provided on the platform for this project. Therefore, assumed strength of 25N/mm² was used for the analysis of the structural elements.

2.0 PURPOSE OF INVESTIGATION

The purpose of the investigation is to:

1. Determine the present status of the structures from the outcome of the visual and Non-Destructive test of the tested structures to ascertain their state with respect to BS EN

12504-4:2021, ASTM C597-09.

2. Determine the ongoing concrete strength of the structure with respect to BS 8110: Part 1 1997.
3. Provide engineering advice based on the visual and Non-Destructive test conducted.
4. To comply with government's statutory requirements.

Recorded tests comprise 2 pulse velocity test(s), 2 crack depth test(s) and 1 surface quality observation(s).

3.0 LITERATURE REVIEW

Ultrasonic Pulse Velocity (UPV) testing is a non-destructive testing technique used for assessing the quality, uniformity, and internal condition of hardened concrete. The method operates by transmitting high-frequency ultrasonic waves through concrete and measuring the travel time between transmitting and receiving transducers.

The pulse velocity is calculated using the relationship:

$$UPV = L / t$$

- L = Distance between transducers (mm)
- t = Pulse transit time (microseconds)

UPV testing is widely used for evaluating concrete homogeneity, detecting internal discontinuities such as cracks and voids, and supporting qualitative assessment of structural integrity.

The testing procedure and interpretation of results are conducted in accordance with international standards including:

- ASTM C597 - Standard Test Method for Pulse Velocity Through Concrete (2020).
- BS EN 12504-4:2004 - Standard Test Method for determination of the velocity of propagation of pulses of ultrasonic longitudinal waves in concrete.
- ACI 228.2R - Report on Nondestructive Test Methods for Evaluation of Concrete in Structures (2018).

Compliance with these standards ensures that equipment calibration, test configuration (direct, semi-direct, or indirect transmission), surface preparation, and reporting procedures meet current international best practice.

It is important to note that pulse velocity is influenced by moisture condition, temperature, aggregate type, and stress level. Therefore, interpretation must consider site conditions and, where necessary, be supported by complementary test methods.

The ultrasonic pulse velocity test is a non-destructive screening method and does not directly determine compressive strength of the reinforced concrete element unless calibrated correlation models are established for the specific concrete mix used in the structure.

Estimated compressive strength (E.C.S) values in this report are derived from the laboratory's fixed

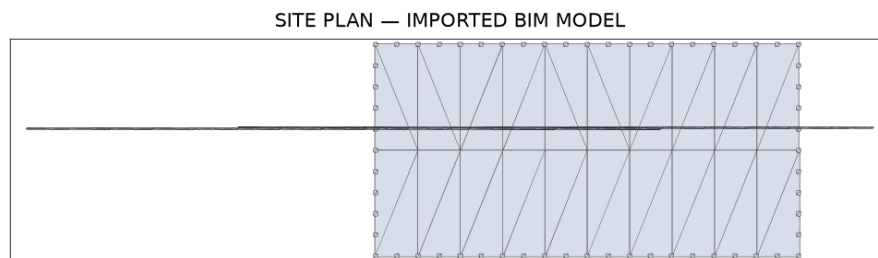
ultrasonic pulse-velocity calibration curve $f_{cu} = 8.961 \times V - 7.97$ (f_{cu} in N/mm², V in km/s), established by least-squares regression over the laboratory's reference control pairs (2.9, 19), (3.9, 25), (4.0, 27), (4.2, 29) and (4.4, 34) and valid over the range 2.0 - 5.0 km/s. BS 1881-203:1999 notes that no unique velocity-strength relationship exists for all concretes; the curve above is the documented calibration applied by this laboratory for Nigerian site concrete and is applied uniformly to every result in Section 5.0. Velocities outside the calibrated range are reported without an E.C.S estimate rather than extrapolated.

3.0 LOCATION MAP/ WEATHER CONDITION

BOTANICAL GARDEN ROAD, EBUTE METTA, LAGOS, LAGOS MAINLAND, LAGOS.

WEATHER CONDITION 31

Recorded concrete surface temperatures: 40.0 deg C.



Plan generated from the project's imported BIM model (PROPOSED STACKING AREA_IFC.rvt, 84 elements) - a plan projection of the model geometry, not a georeferenced survey map.

4.0 FIELD WORK

The dates of the field work were not recorded.

Visual test was carried out on the structure to ascertain any possible structural defects (e.g. cracks, differential settlement, spalling, honeycombs, hogging and sagging). This is a vital aspect of non-destructive test. The Standard Portable Ultrasonic Non-Destructive Digital Indicating Tester (Pundit) was employed for the estimation of compressive strength of the hardened concrete on the structural elements.

It is important to note that during the test some factors were taken into consideration, which may impact on the result of the compressive strength of the structural members. These are as follows:

- Surface conditions, temperature and moisture content of the existing concrete.
- Path length, shape and size of the concrete member.
- Concrete stress.
- Effect of reinforcing bars.

The scope of the work done is as follows:

1. Initial visual test was carried out on the building structure tested (i.e column, beam, slab, wall etc).

2. Indirect method was employed, then between 1 and 3 test points were randomly selected to get a good representation and result on each structural member, according to BS EN 12504-4:2021, for testing concrete.

EQUIPMENT STATUS CHECK

No field device was recorded against these tests.

The test was conducted in the presence of the following recorded operator(s) and staff:

Abdullateef :.....

lateef :.....

lateco :.....

Abdullateef Ojugbele :.....

This is part of Lagos State Government's effort to reduce the incidence of building and civil engineering (construction) materials failure and building collapse within the geographical boundary of Lagos State.

4.1 VISUAL TEST

From the visual inspection conducted on the structure the following observations were noted and recorded as at the time of test;

- a. Floor:200THK RC SLAB:780904: [MANUAL_FIELD_ENTRY - Station UPV-FLD-2026-002] tacky floor
- b. Floor:200THK RC SLAB:781235: [MANUAL_FIELD_ENTRY - Station UPV-FLD-2026-001] observation 1
- c. M_Footing-Rectangular:900 x 900 x 200mm:803486: [MANUAL_FIELD_ENTRY - Station UPV-FLD-2026-002] nigga
- d. M_Footing-Rectangular:900 x 900 x 200mm:803486: ds smooth finishing
- e. M_Footing-Rectangular:900 x 900 x 200mm:803711: [MANUAL_FIELD_ENTRY - Station fghjkjhg] helllo

Following the aforementioned, a Non-Destructive Test was conducted. The photographs in the appendix of this report show the physical state of the structure as at test time.

4.2 METHODOLOGY

NON-DESTRUCTIVE CONCRETE STRENGTH DETERMINATION.

This test is determined by using the Portable Ultrasonic Non-Destructive Digital Indicating Tester (PUNDIT). Non-Destructive, as the name implies, means that the materials being tested are not damaged during the test.

In the Non-Destructive Test, some properties of concrete were measured. These were used to estimate

the strength of the concrete, its elastic behavior and durability, hence determining the integrity of the structural member.

CONCRETE

Pulse velocity measurements made on concrete structures are used for quality control purposes. In comparison with mechanical tests on control samples such as cubes or cylinders, pulse velocity measurements have the advantage that they relate directly to the concrete in the structure rather than to samples, which may not be always truly representative of the concrete in situ.

A pulse of longitudinal vibrations is produced by an electro-acoustical transducer, which is held in contact with one surface of the concrete under test. When the pulse generated is transmitted into the concrete from the transducer using a certified coupling gel material, it undergoes multiple reflections at the boundaries of the different material phases within the concrete. A complex system of stress waves develops, which includes both longitudinal and shear waves, and propagates through the concrete. The first waves to reach the receiving transducer are the longitudinal waves, which are converted into an electrical signal by a second transducer. Electronic timing circuits enable the transit time (T) of the pulse to be measured. This test is conducted for assessing the quality and integrity of concrete by passing ultrasound waves through the specimen under test.

The Pundit test equipment can also determine the following:

- The homogeneity and uniformity of the concrete.
- Changes in the strength of the concrete which may occur with time.
- The quality of the concrete in relation to standard requirements.
- The quality of one element of concrete in relation to another.

Estimated compressive strength conversion: E.C.S: $f_{cu} = 8.961 \times V - 7.97$ (f_{cu} in N/mm², V in km/s; valid 2.0 - 5.0 km/s). The full calibration statement is given in Section 3.0.

4.3 REINFORCING BAR (REBAR) ASSESSMENT

No rebar assessment was recorded during this investigation; the reported results are limited to the ultrasonic and visual indications of Sections 4.1 and 5.0.

4.4 EQUIPMENT/REBAR ASSESSMENT TABLE

No Rebar scanning data recorded for this project.

NOTE: This assessment does not cover for the construction reinforcement design.

Crack depth measurements are tabulated in Section 5.0 (time-difference method).

5.0

ANALYSIS OF TEST RESULT

2 test(s) have not been run through the platform analysis endpoint; velocities shown are computed directly from the recorded measurements using the same deterministic BS 1881-203 relations.

Element average compressive strength is computed from the element mean pulse velocity through the Section 3.0 calibration curve, and members are remarked GOOD or POOR against the statutory 25 N/mm² design strength. Velocities outside the calibrated 2.0 - 5.0 km/s range are reported without an E.C.S estimate rather than extrapolated.

SUMMARY OF TEST ANALYSIS

STRUCTURAL MEMBER	NUMBER TESTED	LOCATION	NO OF POINT TAKEN
M_Footing	1	Floor Not Recorded	1
Floor:200Thk	1	Second Floor	3

FLOOR NOT RECORDED M_FOOTINGS OF BOTANICAL GARDEN ROAD, EBUTE METTA

Structural Element	PATH LENGTH	TRANSIT TIME	PULSE VELOCITY	E.C.S	AVERAGE COMPRESSIVE STRENGTH (N/mm2)	REMARK
M_Footing-Rectangular:900 x 900 x 200mm:803711 (PND-2026-4FAC71)	300	56.0	5.4	-	-	NOT ASSESSED

SECOND FLOOR FLOOR:200THKS OF BOTANICAL GARDEN ROAD, EBUTE METTA

Structural Element	PATH LENGTH	TRANSIT TIME	PULSE VELOCITY	E.C.S	AVERAGE COMPRESSIVE STRENGTH (N/mm2)	REMARK
Floor:200T HK RC SLAB:781235 (PND-2026-F2AC8A)	120	62.5	1.9	-	-	-
-	120	23.7	5.1	-	16.6	POOR
-	120	97.0	1.2	-	-	-

SUMMARY OF TEST RESULTS

STRUCTURAL MEMBER	LOCATION	GOOD (NO, %)	POOR (NO, %)
M_Footing	Floor Not Recorded	0 (0.0%)	0 (0.0%)

STRUCTURAL MEMBER	LOCATION	GOOD (NO, %)	POOR (NO, %)
Floor:200Thk	Second Floor	0 (0.0%)	1 (100.0%)

5.1 CRACK DEPTH MEASUREMENTS (TIME-DIFFERENCE METHOD)

ELEMENT	PATH (MM)	T CRACKED (US)	T UNCRACKED (US)	CRACK DEPTH (MM)	REMARK
Floor:200THK RC SLAB:780904	300.0	70.0	62.5	75.7	Depth exceeds 25 mm - structural review required
M_Footing-Rectangular:900 x 900 x 200mm:803486	200.0	56.0	62.5	-	NOT RECORDED

5.2 SURFACE QUALITY OBSERVATIONS

ELEMENT	SURFACE CONDITION	NOTES
M_Footing-Rectangular:900 x 900 x 200mm:803486	dsmooth finishing	[MANUAL_FIELD_ENTRY - Station UPV-FLD-2026-004] nigga

6.0 RECOMMENDATIONS

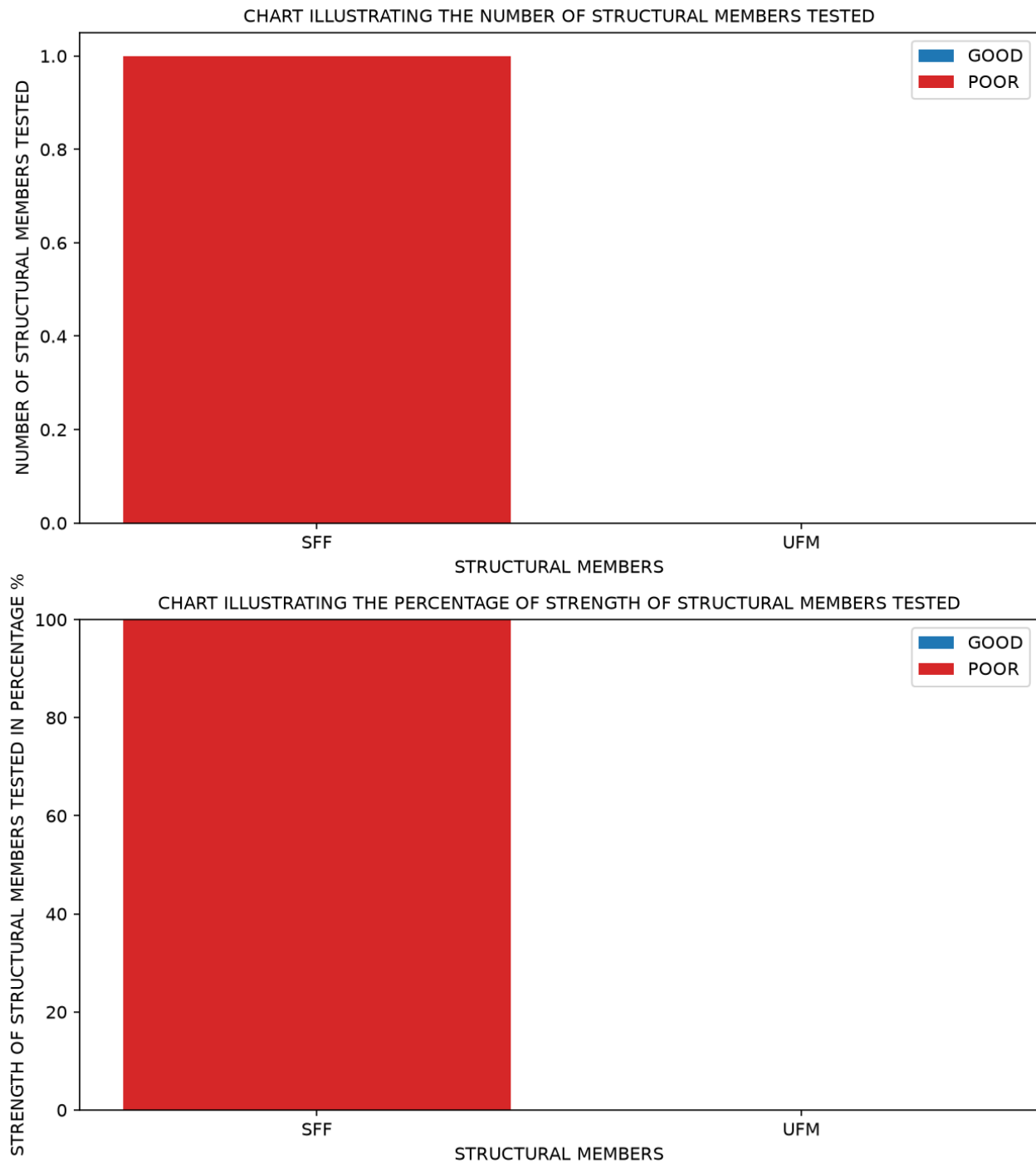
Based on the purpose of investigation, the outcome of the Visual and Non-Destructive Test carried out on the building which shows the present state of the building as described in the visual test and depicted in the bar charts, the test analysis revealed that 0 of the 2 structural members tested in the building were good in strength at the time of test, while 1 fell below the statutory 25 N/mm² strength and require technical advice. It is advised that the client engage a qualified structural engineer and other relevant professionals in the built environment to proffer solution to the defects observed, technical advice on the poor structural members tested and further analyse the structural arrangement to guarantee the stability, integrity and the serviceability of the building.

The platform's stored analysis records add the following recommendations:

- [URGENT] Structural review of 2 element(s) graded poor or very poor by pulse velocity.
- [HIGH] Map and monitor 1 element(s) with crack depth above 25 mm; engage a structural engineer.
- [HIGH] Map crack extent and monitor; engage structural engineer for crack-depth exceedance.
- [URGENT] Immediate structural engineering review - pulse velocity indicates compromised

concrete.

BAR CHART SHOWING SUMMARY OF TEST RESULTS



Legend: SFF: Second Floor Floor:200Thk | UFM: Floor Not Recorded M_Footing

7.0 CONCLUSION

The visual and structural integrity test was conducted in accordance with BS 1881: Part 201: 1986, BS EN 12504-4:2004, BS EN 12504-4:2021.

1. The Non-Destructive Test analysis as shown in the summary of test result (Section 5.0) shows the percentage of strength for the structural elements tested in the building: 0 of 2 elements (0.0%) attained the assumed 25 N/mm² strength at the time of test, while 1 element (50.0%) fell below it.
2. However, it is imperative to state clearly that non-adherence to the recommendation excludes the testing laboratory of any responsibility.

Note: The test assumed 25 N/mm² as the strength of the structural members, however a substructure probe is required to ascertain the integrity of the building foundation.

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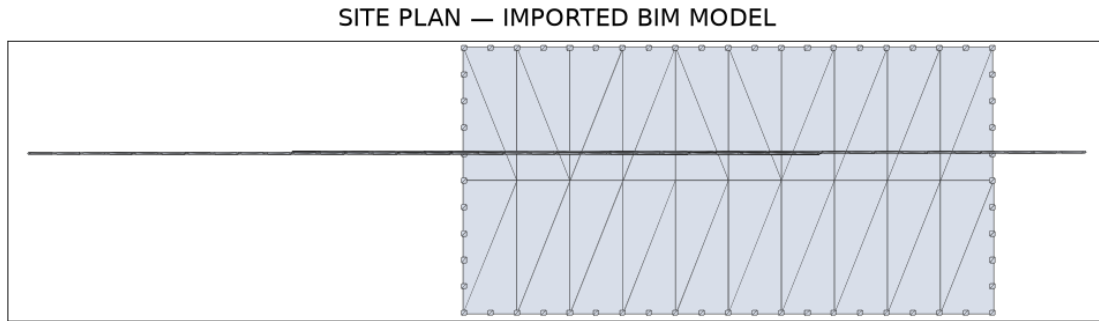
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APPENDIX

DRAWING OF THE BUILDING



Plan generated from the project's imported BIM model (PROPOSED STACKING AREA_IFC.rvt, 84 elements) - a plan projection of the model geometry, not a georeferenced survey map.

PHOTOGRAPHS

PIC I: 1786628996237056000.jpg



PIC II: 1786628911732759040.jpg



PIC III: 1786628988736631808.jpg



PIC IV: 1786628977736079872.jpg



PIC V: rgb image.png



PIC VI: rgb image.png



PIC VII: thermal img.png



PIC VIII: 1786628993736943872.jpg



PIC IX: 1786628998237142784.jpg



PIC X: 1786628996237056000.jpg



PIC XI: 1786628977736079872.jpg

